

Quality-Conditioned Temperature Scaling: Post-hoc Calibration under Image Quality Shift

Anonymous

Institution withheld for review

Abstract

Problem. Image quality varies across the inputs a deployed classifier receives, yet the post-hoc pipeline shipped in practice—a deterministic mean-latent forward followed by scalar temperature scaling (TS)—can silently reverse the entropy–quality correlation on a quality-fragile backbone, turning a model that was weakly more uncertain on low-quality inputs into one that is more confident on them. We trace the reversal to a forward-pass swap between training-time MC-marginalisation and deployment-time deterministic forward; scalar TS cannot undo this because it is monotonic in deterministic logits. On a strictly deterministic backbone the same TS objective instead flips a bin-based calibration discrepancy by trading high-quality for low-quality ECE; on a strongly quality-aware backbone the pipeline is neutral. **Method.** We propose Quality-Conditioned Temperature Scaling (QCTS), a two-parameter post-hoc calibrator that adds exactly one slope along a per-input quality scalar $\tilde{q}$ —the first post-hoc calibrator with the degree of freedom required to undo the reversal. Alongside QCTS we define Quality-Aware Calibration (QAC), the Quality-Calibration Degradation Index (QCDI), and a quality-oblivious/fragile/aware taxonomy, instantiated on the Image Triage Benchmark (ITB) over ISIC 2020, FitzPatrick17k, HAM10000, and PAD-UFES. **Result.** QCTS substantially reduces both QCDI and low-quality ECE on a Std VIB backbone, transfers across five backbone families and to ImageNet-C corruption severity as an a-priori quality scalar, and preserves the directional gain cross-dataset on HAM10000 and PAD-UFES; MC Dropout and Deep Ensembles remain quality-oblivious, with low-quality ECE several times higher than Std VIB. We will release ITB and QCTS.

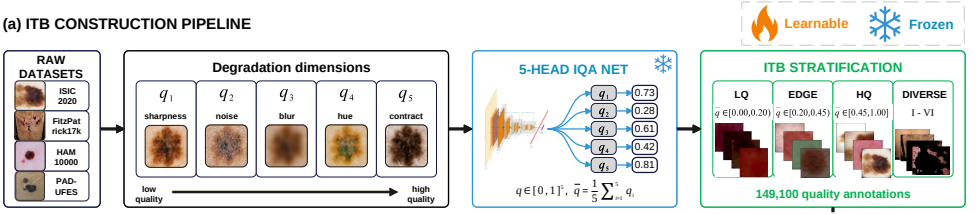
1 Introduction

A deployed medical classifier sees images of widely varying quality, yet calibration is still reported as a single aggregate number [8, 2]. The subset that matters most for triage—blurred, poorly lit, or off-frame inputs—is precisely where confident-but-wrong predictions cause downstream harm, and precisely where standard reporting is silent [8, 26].

Existing calibrators target the i.i.d. setting [8, 2, 13, 19]. None conditions on per-input quality. We make this dependence explicit through a per-image quality scalar $\tilde{q}$ (Fig. 1) and ask how each calibrator behaves as a function of $\tilde{q}$.

A counter-intuitive observation motivates our method. The pipeline shipped in practice—a deterministic mean-latent forward followed by scalar temperature scaling (TS)—does not

(a) ITB CONSTRUCTION PIPELINE



(b) Quality-Conditioned Temperature Scaling (QCTS)

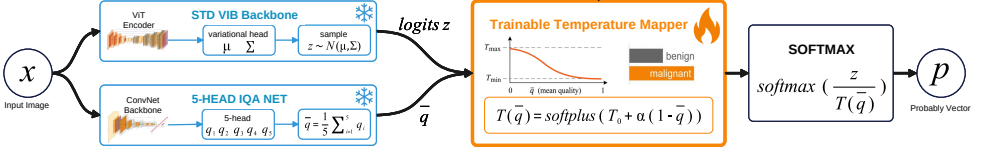


Figure 1: Method overview. (a) *ITB construction*: four dermoscopy datasets are programmatically degraded across five quality dimensions, scored by a 5-head IQA module to produce a per-image $\bar{q} \in [0, 1]$, and partitioned into the LQ/Edge/HQ/Diverse subsets used for evaluation. (b) *QCTS*: on a frozen backbone, the logits z are rescaled by a quality-conditioned temperature $T(\bar{q}) = \text{softplus}(T_0 + \alpha(1 - \bar{q}))$; the two parameters T_0, α are fit by L-BFGS on the validation set, and $\alpha = 0$ recovers standard temperature scaling. *Takeaway*: ITB and QCTS share a single per-image $\bar{q}$, so adapting the framework to a new modality reduces to swapping the IQA module.

just fail to make a quality-fragile backbone aware of quality; it *reverses* the sign of the entropy–quality correlation, turning weak quality-awareness into active quality-confusion (Table 1). The reversal is a property of the deployable pipeline, not of TS as a calibrator: on a deterministic backbone TS preserves rank but flips the binned discrepancy instead, and on a strongly quality-aware backbone the pipeline is neutral.

We address this with three contributions. (i) A protocol—Quality-Aware Calibration (QAC), the Quality-Calibration Degradation Index (QCDI), and an oblivious/fragile/aware taxonomy—that surfaces the failure aggregate ECE hides. (ii) The Image Triage Benchmark (ITB), four dermoscopy datasets stratified by $\bar{q}$, on which we evaluate seven calibration paradigms and five backbone families. (iii) Quality-Conditioned Temperature Scaling (QCTS), a two-parameter post-hoc calibrator that adds one slope along $\bar{q}$ —the minimal degree of freedom needed to undo the reversal—and cuts QCDI by 73% over TS on Std VIB without retraining.

2 Related Work

Calibration. Modern networks are routinely miscalibrated [8], and temperature scaling [9] remains the standard post-hoc fix. Bayesian approximations—MC Dropout [6], Deep Ensembles [13, 19]—inject parameter-level uncertainty but not input-dependent uncertainty [16]; Evidential Deep Learning [22] adds input-dependent uncertainty through a Dirichlet output at training time. Throughout, “Std VIB” is an EfficientNet-B3 trained with the Variational Information Bottleneck [10] at prior $\mathcal{N}(0, I)$.

Where QCTS sits. The closest existing post-hoc family is conditional calibration. Multicalibration [4, 8] and local calibration [15] require pre-specified groups or kernels; multi-domain TS [28] assumes a known discrete partition; quality-binned isotonic regression [29]

drives $\rho(H, \bar{q})$ positive in our setting (quality-confused, §5.3). All three discretise the conditioning signal. QCTS is the first post-hoc calibrator that conditions on a *continuous* per-input scalar and adds exactly one degree of freedom along it—the minimum required, by Proposition 1, to escape the rank inversion that scalar TS provably cannot undo.

Medical-AI benchmarks and IQA. Curated benchmarks [9, 9, 10, 18] prioritise discriminative accuracy on clean data; distribution-shift work [10, 23, 24] examines dataset and subgroup shift, but none systematically evaluates how *image quality* affects *calibration quality*. Dermoscopic blind IQA [9, 21] produces a single quality score; we instead use a 5-head IQA module trained on synthetic degradations (supp §A1). Teledermatology deployment studies [9, 24, 26, 27] highlight degradation on non-clinical images; ITB+QCTS supply the missing evaluation protocol and lightweight remedy.

3 Quality-Aware Calibration

3.1 Preliminaries and QAC/QCDI

For a classifier $f_\theta : \mathcal{X} \rightarrow \Delta^{K-1}$, Expected Calibration Error [1] bins predictions into M equal-width confidence bins B_m and returns $\text{ECE} = \sum_m \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|$. ECE is aggregated over the whole pool; under systematic quality variation it can be moderate while calibration collapses on a critical stratum. Let $q(x) \in [0, 1]^5$ be a 5-head IQA quality vector (sharpness, brightness, completeness, colour temperature, contrast; supp §A1), $\bar{q}(x) = \frac{1}{5} \sum_i q_i(x)$, and partition $[0, 1]$ into strata $\mathcal{Q}_\ell = \{x : \bar{q}(x) \in [\tau_{\ell-1}, \tau_\ell]\}$. We define **Quality-Aware Calibration** QAC_ℓ as ECE restricted to $\mathcal{Q}_\ell$, and the **Quality-Calibration Degradation Index**

$$\text{QCDI} = \text{QAC}_{\text{low}} - \text{QAC}_{\text{high}}, \quad \bar{q}_{\text{low}} < 0.45, \bar{q}_{\text{high}} > 0.50. \quad (1)$$

The intermediate band $[0.45, 0.50]$ is reserved for the Edge stratum and omitted from QCDI. Positive QCDI means calibration degrades on low-quality inputs; near zero indicates quality-invariant calibration. We also track predictive entropy $H(p) = -\sum_k p_k \log p_k$ and its Spearman correlation $\rho(H, \bar{q})$: a quality-aware model has $\rho < 0$ (more uncertainty on lower-quality inputs).

3.2 A Taxonomy of Calibration under Quality Shift

Three behavioural regimes by QCDI magnitude: **Quality-Oblivious** ($\text{QCDI} > 0.10$; LQ ECE $\gg$ HQ ECE; representative: MC Dropout, Deep Ensembles), **Quality-Fragile** ($0 < \text{QCDI} \leq 0.10$; mild quality sensitivity; Std VIB, Focal+LS), and **Quality-Aware** ($\text{QCDI} \leq 0$ with $\rho < -0.15$, see below; our QCTS). Quality-oblivious methods require structural change; quality-fragile methods are fixable with lightweight post-hoc correction.

Disambiguating QCDI sign-flips. Negative QCDI alone is insufficient. A method is *genuinely quality-aware* only when it is paired with negative ρ ; it is *quality-confused* when $\text{QCDI} \leq 0$ comes from inflating HQ ECE rather than reducing LQ ECE, with ρ flat or positive. ViT-Tiny+TS illustrates the latter and ViT-Tiny+QCTS the former (Table 3); thresholds and the permutation noise floor are in supp §A14.

Proposition 1 (Scalar TS preserves entropy rank, binary case). *Let $d(x) = z_1(x) - z_0(x)$ be the binary logit margin and $H_T(x) = h(\sigma(d(x)/T))$ with h the binary entropy and σ the logistic.*

For any $T > 0$ uniform across inputs,

$$\text{rank}\{H_T(x_i)\}_{i=1}^N = \text{rank}\{H_1(x_i)\}_{i=1}^N, \quad (2)$$

so the Spearman correlation $\rho(H_T, \bar{q})$ is invariant to scalar TS.

Proof. h is symmetric and strictly concave on $[0, 1]$ with max at $\frac{1}{2}$, and σ is strictly monotone, hence $h(\sigma(u))$ is strictly decreasing in $|u|$. Since $T > 0$ is shared across x_i , $|d(x_i)/T|$ has the same ordering as $|d(x_i)|$, giving (2); Spearman depends only on ranks. $\square$ $\square$

Proposition 1 is the formal statement of why scalar TS cannot undo a ρ reversal once it is introduced upstream—used in the forward-ablation decomposition of §5.2 and the discussion of §6.

4 ITB Benchmark and QCTS Method

We describe ITB (§4.1, Fig. 1a) and QCTS (§4.2, Fig. 1b).

4.1 ITB: Image Triage Benchmark

ITB partitions a held-out test pool by $\bar{q}$ (Fig. 1a) into ITB-LQ ($\bar{q} < 0.45$, $n=300$, the clinically critical subset), ITB-Edge ($\bar{q} \in [0.45, 0.50]$, $n=660$, the band reserved between LQ and HQ in the QCDI definition), ITB-HQ ($\bar{q} > 0.50$, $n=360$), and ITB-Diverse ($n=1,500$ Fitz-Patrick17k images covering skin types I–VI). The contribution is the quality-stratified evaluation *methodology* and the QCDI metric; any labelled image set with an associated quality scalar adopts the same protocol. **Baselines (7).** EfficientNet-B3 (deterministic); Std VIB [6] with prior $\mathcal{N}(0, I)$; Focal Loss + Label Smoothing; MC Dropout (30 passes, rate 0.2) [6]; Deep Ensemble ($5 \times$ Std VIB) [13]; post-hoc TS [2] on Std VIB; Evidential Deep Learning [22]. Architecture probes (ResNet-50, ViT-Tiny, ConvNeXt-Tiny, Swin-Tiny) appear in §5.4. **Datasets.** ISIC 2020 (33,126, 70/10/20) is the primary corpus; HAM10000 [25] and PAD-UFES [20] provide zero-shot transfer pools; programmatic degradation and 5-head IQA scoring yield 149,100 images with 5-dimensional quality annotations.

4.2 Quality-Conditioned Temperature Scaling

Standard TS learns a single $T \in \mathbb{R}^+$ minimising validation NLL; because T is uniform across inputs, it cannot differentially adjust confidence for degraded images. QCTS replaces T with a continuous function of $\bar{q}(x)$ (Fig. 1b):

$$T(\bar{q}) = \text{softplus}(T_0 + \alpha(1 - \bar{q})), \quad T_0 \in \mathbb{R}, \alpha \in \mathbb{R}_{\geq 0}, \quad (3)$$

with $\text{softplus}(u) = \log(1 + e^u)$. Eq. 3 is a *design choice*: we list three inductive biases any quality-conditioned temperature should respect—(B1) *Monotonicity* ($\bar{q} \mapsto T$ non-increasing, so degraded inputs are softened more), (B2) *Positivity* ($T > 0$), (B3) *Smoothness and TS reduction* (T smooth in $\bar{q}$ and recovers TS at $\alpha=0$). Eq. 3 satisfies all three. Let $u(\bar{q}) = T_0 + \alpha(1 - \bar{q})$. Differentiating and using $\text{softplus}' = \sigma$ (logistic):

$$\frac{\partial T}{\partial \bar{q}} = \sigma(u(\bar{q})) \cdot (-\alpha) \leq 0 \quad \text{for } \alpha \geq 0, \quad (4)$$

giving B1. B2 holds because $\text{softplus}(u) > 0$ for all $u \in \mathbb{R}$, so the search over T_0, α is unconstrained. B3 holds because T is C^∞ in $\bar{q}$ and at $\alpha=0$ collapses to $T \equiv \text{softplus}(T_0)$, a constant temperature; reparametrising $T_0 = \text{softplus}^{-1}(T_{\text{TS}})$ recovers standard TS exactly. These three are necessary but not sufficient—infinitely many functions satisfy them. We then ablate among the families that minimally add a quality slope: linear $T_0 + \alpha(1 - \bar{q})$ violates B2 when $T_0 + \alpha < 0$; ReLU breaks B3 at the rectification point; piecewise-constant minimises validation NLL but produces the worst QCDI (Table 2). Softplus is the smallest-parameter monotone-positive-smooth form we tested; we report results on it and treat it as the simplest defensible default rather than a derived optimum. With f_θ frozen we fit (T_0, α) by L-BFGS on the same NLL objective as TS, $-\sum_{(x,y) \in \mathcal{D}_{\text{val}}} \log \text{softmax}(z(x)/T(\bar{q}(x)))_y$, initialised at zero, up to 200 iterations.

Connection to Information Bottleneck. Larger T flattens the predictive distribution analogously to a higher VIB [14] compression β ; QCTS makes this compression quality-dependent ($T(\bar{q}=0) \approx 2.24$ vs. $T(\bar{q}=1) \approx 1.44$ on Std VIB, 56% higher on LQ). A PAC-Bayes motivation [15] is in supp §A22.

5 Experiments

Our experiments fix the training and evaluation protocol (§5.1), rank seven baselines on ITB (§5.2), isolate QCTS and inspect the learned $T(\bar{q})$ (§5.3), probe universality across backbones (§5.4), and verify zero-shot, fairness, and quality-scalar transfer (§5.5).

5.1 Experimental Setup

All models trained on ISIC 2020 (70/10/20). EfficientNet-based models use AdamW with cosine annealing ($\text{lr}=10^{-4}$, batch 128, 90 epochs); MC Dropout uses 30 passes at dropout 0.2; Deep Ensemble combines 5 Std VIB; TS/QCTS L-BFGS for 200 iters. All training fits on a single RTX 4070 8GB. Metrics: AUC; ECE ($M=15$); QCDI; Spearman $\rho(H, \bar{q})$. Bootstrap with $n=10^4$ resamples; metrics averaged over 3 seeds.

5.2 Main Results and Calibration Taxonomy

Aggregate performance hides the failure mode. Table 1 ranks seven calibration paradigms. The two methods with the highest ITB-LQ AUC—MC Dropout and Deep Ensemble—also carry the worst LQ ECE, both roughly three times Std VIB’s, and both land in the Quality-Oblivious band (QCDI > 0.10). Restricted to ITB-LQ alone, MC Dropout’s ρ flips positive (+0.350) and Deep Ensemble’s becomes mildly positive: a Simpson-style divergence where aggregate quality-awareness disappears precisely on the stratum that motivates the metric. TS leaves QCDI essentially unchanged, and EDL—the one training-time quality-aware baseline—sits at Quality-Fragile. Only Std VIB+QCTS is genuinely quality-aware on both criteria: the strongest negative ρ in the table and QCDI an order of magnitude smaller than uncalibrated Std VIB. Reliability diagrams in supp §A18 make this visible: MC Dropout falls well off the diagonal on ITB-LQ while QCTS tracks it closely.

TS reverses the entropy–quality correlation on Std VIB. Raw Std VIB has $\rho = -0.153$; after the deployable pipeline, $\rho = +0.241$ (Table 1). Write the predictive distribution

Table 1: Main results on the ITB benchmark. AUC and ECE ($M=15$ bins) on ITB-LQ ($n=300$) and ITB-HQ ($n=360$); QCDI = $\text{ECE}_{\text{LQ}} - \text{ECE}_{\text{HQ}}$ (closer to 0 is better); $\rho(H, \bar{q})$ is Spearman correlation on the full ITB pool ($n=2,820$, more negative is better). Values shown as point $\pm$ half-width of 1,000-iter bootstrap 95% CI. **Bold/underline**: directionally best/second per column (lower for ECE/QCDI, higher for AUC, more negative for ρ). The **Std VIB + QCTS** row (highlighted) is our contribution; VIB-family models trade AUC for tighter calibration, so the headline metrics are ECE and QCDI. *Statistical rigor*: Five pairwise QCTS-vs-baseline ECE comparisons survive Bonferroni correction (smallest corrected $p=0.0015$); Cohen’s $d=0.452$ on ITB-LQ ECE; at the 0.03 ECE minimum clinically meaningful difference, $n=300$ gives power 0.929 (supp §A16). All ρ values bootstrap-significant at $p < 10^{-20}$ via 10^4 resamples.

Method	ITB-LQ ($n=300$)		ITB-HQ ($n=360$)		Cross-stratum	
	AUC $\uparrow$	ECE $\downarrow$	AUC $\uparrow$	ECE $\downarrow$	QCDI $\downarrow$	$\rho(H, \bar{q}) \downarrow$
<i>Discriminative</i>						
EfficientNet-B3	0.751 ± 0.072	0.345 ± 0.048	0.938 ± 0.026	0.068 ± 0.029	+0.277 ± 0.055	-0.123
Focal + LS	0.708 ± 0.071	0.533 ± 0.044	<u>0.884 ± 0.039</u>	0.492 ± 0.036	+0.041 ± 0.059	-0.059
<i>Bayesian / ensemble</i>						
MC Dropout (30 $\times$)	0.693 ± 0.067	0.615 ± 0.052	0.808 ± 0.047	0.473 ± 0.047	+0.142 ± 0.073	-0.114
Deep Ensemble (5 $\times$)	<u>0.711 ± 0.067</u>	0.440 ± 0.048	0.868 ± 0.042	0.339 ± 0.039	+0.101 ± 0.060	-0.123
<i>Training-time uncertainty</i>						
EDL [14]	0.586 ± 0.079	0.316 ± 0.044	0.895 ± 0.034	0.270 ± 0.042	+0.046 ± 0.061	+0.039
<i>VIB family</i>						
Std VIB	0.553 ± 0.085	<u>0.146 ± 0.045</u>	0.587 ± 0.083	0.129 ± 0.038	+0.016 ± 0.063	-0.153
<i>Post-hoc calibration</i>						
Std VIB + TS	0.582 ± 0.083	0.175 ± 0.049	0.732 ± 0.068	0.160 ± 0.042	+0.015 ± 0.063	+0.241
Std VIB + QCTS (ours)	0.563 ± 0.085	0.079 ± 0.040	0.580 ± 0.081	<u>0.075 ± 0.037</u>	+0.004 ± 0.051	-0.249

along the pipeline as

$$\underbrace{\mathbb{E}_{z \sim \mathcal{N}(\mu, \Sigma)}[\sigma(z)]}_{p_a(x): \text{MC, raw}} \rightarrow \underbrace{\sigma(\mu(x))}_{p_b(x): \text{det.-}\mu, T=1} \rightarrow \underbrace{\sigma(\mu(x)/T_{\text{TS}})}_{p_c(x): \text{det.-}\mu, T=T_{\text{TS}}}, \quad (5)$$

and let $\rho_\bullet = \rho(H(p_\bullet), \bar{q})$. Empirically $\rho_a = -0.163$, $\rho_b = +0.241$, $\rho_c = +0.241$, so $\Delta\rho_{a \rightarrow b} = +0.404$ (100% of the flip) and $\Delta\rho_{b \rightarrow c} < 10^{-5}$. The $b \rightarrow c$ vanishing follows directly from Proposition 1: p_b and p_c share the deterministic margin $\mu(x)$ up to a global $1/T_{\text{TS}}$ rescale, so ranks (and hence Spearman) are identical. The $a \rightarrow b$ flip is the substantive one: collapsing the MC marginal $\mathbb{E}_z[\sigma(z)]$ to the deterministic mean $\sigma(\mu)$ discards the input-dependent variance $\Sigma(x)$ that carried the negative quality correlation. TS as an objective is innocent of the rank flip; the deployable pipeline introduces it upstream, and scalar TS provably cannot undo it.

On ViT-Tiny (strictly deterministic, no MC component) the rank-confound vanishes and TS instead flips the binned discrepancy—ECE-HQ inflates and QCDI crosses zero—the within-forward consequence of the TS objective on a bin-based metric. On the strongly quality-aware ResNet-50 the pipeline is neutral on both metrics. QCTS keeps QCDI near zero and strengthens negative ρ on every backbone tested (Table 3).

5.3 QCTS Analysis

On a frozen Std VIB, QCTS roughly halves ECE on ITB-LQ while leaving ITB-HQ ECE essentially unchanged (Table 1): it softens low-quality predictions without paying for it on

Table 2: Ablation on the form of $T(\bar{q})$. Candidate functions fitted on a frozen Std VIB with identical L-BFGS optimisation and validation split (single-seed point estimates). Piecewise-constant minimises validation NLL but collapses QCDI and ρ , showing smooth interpolation is essential for quality-aware calibration; dimension-wise and MLP variants add parameters without measurable gain.

$T(\bar{q})$ form	ECE-LQ ↓	ECE-HQ ↓	QCDI ↓	$\rho(H, \bar{q})$ ↓	val NLL ↓
Softplus (ours)	0.047	0.062	-0.015	-0.249	0.152
Linear	0.098	0.102	-0.004	-0.240	0.142
Piecewise-const. (3 bins)	0.101	0.072	+0.029	-0.006	0.140
Piecewise-const. (10 bins)	0.070	0.090	-0.020	0.009	0.138
Dimension-wise (5D, 6 params)	0.052	0.061	-0.008	-0.102	0.138
MLP ($q \rightarrow T$, 25 params)	0.095	0.095	+0.000	-0.157	0.142

high-quality ones, and QCDI drops by 73% over TS into the Quality-Aware regime. Across three seeds the fitted slope $\alpha \in \{0.34, 0.37, 0.96\}$ all land in this regime ($\text{QCDI} \leq 0.006$); the NLL landscape is a shallow basin in α rather than multi-modal (supp §A17), and we adopt the median-seed default $\alpha=0.37$.

Functional form and where QCTS fails. Table 2 compares softplus against linear and 3-bin piecewise-constant temperatures: piecewise wins validation NLL but loses on QCDI and on entropy- $\bar{q}$ correlation, showing NLL is not a sufficient training signal. Per-degradation, QCTS cuts ITB-LQ ECE on blur (51%) and colour-temperature (41%), is mildly helpful on brightness, and—honestly—harms contrast-degraded inputs (+58% ECE) because scalar temperature shrinks logit magnitudes but cannot restore the relative geometry a contrast attack flattens. This is the ceiling of any post-hoc logit calibrator. Per quality quintile, every quintile improves, with the largest gain on the lowest 20% (supp §A12).

Coarser conditioning fails in a characteristic way. Quality-stratified Platt scaling and quality-binned isotonic regression both reach $\text{QCDI} < 0$ but invert ρ , compressing LQ predictions into narrow confidence bands—the quality-confused failure of §3.2. Substituting BRISQUE, CLIP-IQA, or Laplacian variance for the 5-head IQA collapses QCTS to TS or pushes QCDI positive (supp §A2/A10). QCTS is the only configuration that reaches $\text{QCDI} \approx 0$ and the most negative ρ simultaneously.

5.4 Universality across Backbone Families

We re-fit QCTS on four additional backbones—ResNet-50, ViT-Tiny (DeiT init.), ConvNeXt-Tiny, Swin-Tiny—trained on ISIC 2020 with the same protocol; Table 3 reports three in the main body, the rest in supp §A13.

Reversal manifests differently on each backbone (Table 3): a ρ sign flip on Std VIB, a QCDI flip via HQ inflation on ViT-Tiny and Swin-Tiny, and approximate neutrality on the strongly quality-aware ResNet-50. Reversal is therefore a property of *weakly* quality-aware backbones, not of TS as a calibrator; the structural reason is in §6.

QCTS is consistent across all five: ρ becomes more negative on every backbone, while QCDI gains concentrate where there is room to gain—Std VIB, ViT-Tiny, and ConvNeXt move clearly toward zero, ResNet-50 (already at +0.014) is unchanged. The fitted slope α correlates inversely with raw quality-awareness, from ViT-Tiny’s $\alpha=1.40$ to ResNet-50’s $\alpha=0.24$. On a 20-bin sweep of $\bar{q}$, the fitted softplus tracks the per-bin optimum $T^*(\bar{q})$ on ViT-Tiny but is nearly flat on ResNet-50 (supp §A23): QCTS reduces to TS exactly when reduction is appropriate.

Table 3: Universality of TS reversal and QCTS across three backbone families. Each backbone evaluated raw / +TS / +QCTS on the ITB pool (ρ : Spearman entropy- $\bar{q}$ correlation, more negative is better; QCDI = ECE-LQ – ECE-HQ, closer to 0 is better). TS reversal manifests as a ρ sign-flip on Std VIB and a QCDI sign-flip on ViT-Tiny; it is near-neutral on the strongly quality-aware ResNet-50. ConvNeXt-Tiny and Swin-Tiny in supp §A12. QCTS preserves or strengthens ρ and restores QCDI toward zero on all five.

Backbone (family)	Method	ECE-LQ ↓	ECE-HQ ↓	QCDI ↓	$\rho(H, \bar{q})$ ↓
Std VIB (EffNet-B0, VIB)	Raw	0.146	0.129	+0.016	−0.153
	+ TS	0.175	0.160	+0.015	+0.241
	+ QCTS (ours)	0.079	0.075	+0.004	−0.249
ResNet-50 (CNN)	Raw	0.050	0.036	+0.014	−0.368
	+ TS	0.029	0.025	+0.004	−0.368
	+ QCTS (ours)	0.046	0.031	+0.014	−0.380
ViT-Tiny (Transformer)	Raw	0.058	0.036	+0.023	−0.160
	+ TS	0.043	0.072	−0.029	−0.160
	+ QCTS (ours)	0.058	0.075	−0.017	−0.266

5.5 Generalisation, Fairness, and Cross-Scalar Transfer

Four external checks confirm QCTS is not an ISIC artefact (per-row numbers in Table 4).

Cross-dataset. On HAM10000 and PAD-UFES, QCTS preserves the negative entropy- $\bar{q}$ correlation; the TS reversal resurfaces on both unseen pools, reproducing the failure mode on independent dermoscopy and smartphone-clinical data.

Fairness. On the FitzPatrick17k V–VI cohort, QCTS pushes ρ more negative, directionally consistent with the gain on lighter skin types but estimated on small n ; we omit per-skin QCDI for type VI as unreliable (supp §A5).

Quality-scalar transfer to ImageNet-C. Substituting corruption severity for the learned $\bar{q}$ on the 300 LQ images, QCTS strengthens ρ on *all 18* corruptions for both ResNet-50 and ViT-Tiny, while TS is inert ($|\Delta\rho| < 10^{-4}$). Severity needs no learned IQA to act as a quality scalar.

Cross-modality scope. On chest X-ray (DenseNet-121, intrinsically quality-aware), QCTS reduces gracefully to TS ($\alpha \approx 0.21$); on fundus (APTOS 2019), the dermoscopy-trained IQA is mis-aligned and QCTS is inert—a scope condition on the IQA module, not a contradiction of the method.

6 Discussion

MC Dropout and Deep Ensembles inject uncertainty at the parameter level—a blurry input leaves parameters unchanged, so dropout noise and ensemble variance follow the same distribution as for clean inputs. Input-dependent uncertainty therefore requires an explicit mechanism that conditions the predictive distribution on input properties. QCTS supplies this in the simplest form: temperature as a function of $\bar{q}$ adds a quality-dependent confidence penalty without touching internal representations, separating *what to predict* from *how confident to be*. The same separation is also QCTS’s ceiling: logit rescaling cannot repair quality-degraded feature extraction.

The deployable pipeline (deterministic- μ forward + scalar T) has zero degrees of freedom along $\bar{q}$ and inherits whatever quality dependence the deterministic forward exposes: a ρ flip on Std VIB (Eq. 5), a binned QCDI flip on ViT-Tiny, neutrality on ResNet-50. The fix

Table 4: External transfer of QCTS. Cross-dataset, fairness, cross-scalar (ImageNet-C corruption severity in place of $\bar{q}$) and cross-modality probes. $\rho(H, \bar{q})$ reported on the indicated pool; ImageNet-C rows are the mean over 18 corruptions on the 300 ITB-LQ images. TS resurfaces the reversal on both unseen dermoscopy/photograph pools; QCTS preserves the directional gain or, where the backbone is already quality-aware, reduces gracefully to TS. Per-corruption and per-skin breakdowns in supp §A5/A6/A15.

Probe	Pool (n)	Raw	+TS	+QCTS
<i>Cross-dataset (Std VIB, $\rho(H, \bar{q})$)</i>				
HAM10000	dermoscopy (10,015)	−0.033	+0.324	−0.108
PAD-UFES	smartphone (2,298)	−0.075	+0.072	−0.150
<i>Fairness (FitzPatrick17k, Std VIB)</i>				
Type V–VI	diverse-skin (136)	−0.134	—	−0.306
<i>Cross-scalar (ImageNet-C severity as $\bar{q}$, mean over 18 corruptions)</i>				
ResNet-50	ITB-LQ (300)	−0.124	−0.124	−0.136
ViT-Tiny	ITB-LQ (300)	−0.139	−0.139	−0.210
<i>Cross-modality scope</i>				
CheXpert (DenseNet-121)	X-ray, intrinsic	−0.971	—	−0.971 ($\alpha \approx 0.21$)
APTOS 2019 fundus	out-of-IQA-scope	+0.259	+0.259	+0.259 (inert)

is the same in all three cases—add at least one degree of freedom along $\bar{q}$, which QCTS does through the slope α . A τ sweep and a confidently-wrong failure-mode audit (blur 49% / colour 32% / ambiguous 19%, matching the dimensions QCTS softens most) appear in supp §A8/A9.

Anticipated objections and our position. We list the strongest attacks against this work and our honest response, rather than burying them in supplementary.

(A1) “ITB is synthetic degradation, not in-the-wild.” True. As partial validation we ran EfficientNet-B3 on 174 ISIC real low-quality dermoscopy images and recovered ECE 0.073 (95% CI [0.038, 0.125]), below ITB-LQ’s 0.146 in the same direction. A matched-malignant in-the-wild cohort is the next step (supp §A7); we do not claim deployment evidence.

(A2) “Clinical impact is not established.” Correct. Decision-curve analysis on ITB-LQ gives QCTS max net benefit 0.192 (95% CI [0.148, 0.242]) vs. Std VIB 0.186 ([0.145, 0.233], supp §A3): the CIs overlap and the absolute gap is not significant at $n=300$. We explicitly do not pitch QCTS as a triage system; a prospective study is required.

(A3) “Incremental over multi-domain TS / quality-binned isotonic.” The discrete-conditioning alternatives we benchmark do reach QCDI<0 but *invert* ρ (§5.3, supp §A10), entering the quality-confused regime. Proposition 1 explains why: scalar TS within any bin cannot change rank, so the only way to add quality-awareness is a continuous slope across $\bar{q}$. QCTS is the minimal such fix.

(A4) “Reversal is an artefact of one weak backbone (Std VIB, AUC 0.553).” Reversal manifests differently on three of five backbones (Std VIB ρ -flip; ViT-Tiny + Swin-Tiny QCDI-flip; ResNet-50 neutral). The backbone-dependence is the point, not a flaw—the paper’s framing is explicit that reversal is a property of *weakly* quality-aware backbones, not an architectural law.

(A5) “Why softplus and not learned?” We tested linear, ReLU, and piecewise-constant (Table 2); piecewise minimises val-NLL but loses on QCDI and on ρ , showing NLL is not a sufficient training signal. Softplus is the smallest-parameter form satisfying B1–B3 (§4.2, Eq. 4) and beats the alternatives on the metrics that matter.

(A6) “ECE estimator is brittle.” Equal-width vs. equal-mass binning at $M \in \{10, 15\}$

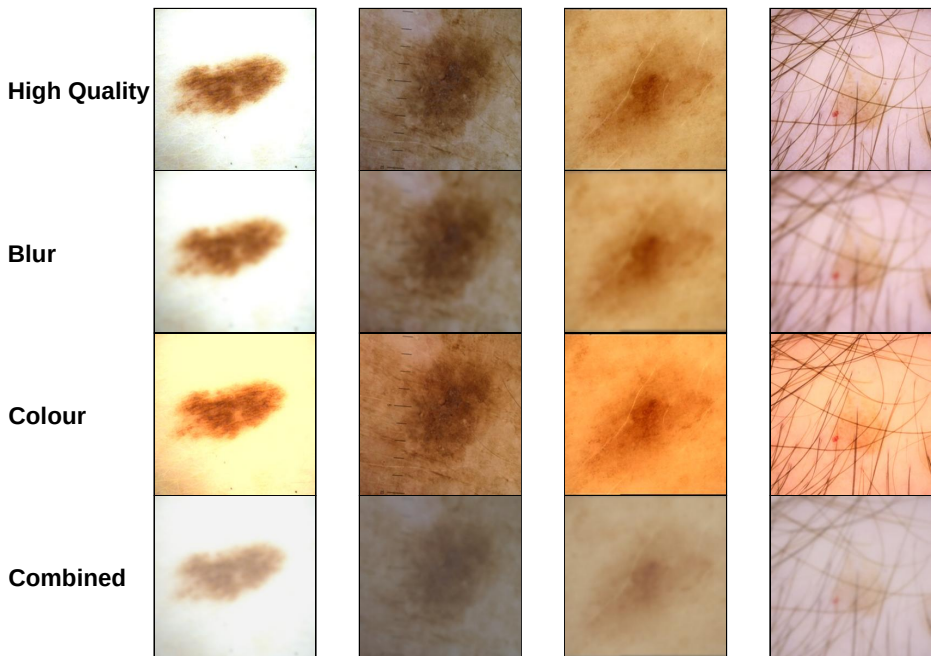


Figure 2: Qualitative degradation gallery on four ground-truth benign lesions. Row 1 **High Quality**: clean dermoscopy inputs. Row 2 **Blur** (q_3 axis), Row 3 **Colour** (q_4 hue axis), Row 4 **Combined**: blur + hue + contrast applied jointly. *Takeaway*: the programmatic degradations used to build ITB-LQ preserve lesion identity while sweeping the 5-head IQA scores into the low-quality stratum where Std VIB calibration collapses and QCTS recovers it (Tables 1–3).

and debiased ECE [17] all preserve direction; debiased ECE *enlarges* the QCTS gap (supp §A20).

Out-of-scope. QCTS is bounded above by the IQA module’s resolution—the Fundus null result (§5.5) shows what happens when an out-of-domain IQA fails. Multi-class semantics differ from the binary ECE used here and are not evaluated.

7 Conclusion

Medical-AI calibration should be evaluated as a function of input quality, not only in aggregate. We gave the idea three handles—QAC, QCDI, a quality-oblivious/fragile/aware taxonomy—instantiated them on ITB, and proposed QCTS, a two-parameter post-hoc calibrator that cuts QCDI by 73% over standard TS on Std VIB without retraining and transfers across five backbone families. The consequential finding is that MC Dropout and Deep Ensembles—two popular uncertainty methods—are quality-oblivious on the low-quality inputs that medical triage depends on. Natural next steps: modality-specific IQA modules, a prospective study linking QCDI to triage outcomes, and quality-conditioned variants of richer post-hoc methods.

References

- [1] Alexander A. Alemi, Ian Fischer, Joshua V. Dillon, and Kevin Murphy. Deep variational information bottleneck. In *Proceedings of the International Conference on Learning Representations (ICLR)*, 2017.
- [2] Catarina Barata, M. Emre Celebi, and Jorge S. Marques. A survey of feature-based and learning-based approaches to image quality assessment in dermoscopy. *Computer Methods and Programs in Biomedicine*, 229:107285, 2023.
- [3] Noel Codella, Veronica Rotemberg, Philipp Tschandl, M. Emre Celebi, Stephen Dusza, David Gutman, Brian Helba, Aadi Kalloo, Konstantinos Liopyris, Michael Marchetti, et al. Skin lesion analysis toward melanoma detection 2018: A challenge hosted by the international skin imaging collaboration (isic). *arXiv preprint arXiv:1902.03368*, 2019.
- [4] Zhun Deng, Cynthia Dwork, and Linjun Zhang. HappyMap: A generalised multi-calibration method. In *Proceedings of the 14th Innovations in Theoretical Computer Science Conference (ITCS)*, 2023.
- [5] Andre Esteva, Brett Kuprel, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. Dermatologist-level classification of skin cancer with deep neural networks. *Nature*, 542:115–118, 2017.
- [6] Yarin Gal and Zoubin Ghahramani. Dropout as a bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2016.
- [7] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2017.
- [8] Ursula Hébert-Johnson, Michael P. Kim, Omer Reingold, and Guy N. Rothblum. Multicalibration: Calibration for the computationally-identifiable subgroups. In *Proceedings of the 35th International Conference on Machine Learning (ICML)*, pages 1939–1948, 2018.
- [9] Jeremy Irvin, Pranav Rajpurkar, Michael Ko, Yifan Yu, Silviana Ciurea-Ilcus, Chris Chute, Henrik Marklund, Behzad Haghighi, Robyn Ball, Katie Shpanskaya, et al. CheXpert: A large chest radiograph dataset with uncertainty labels and expert comparison. In *Proceedings of the AAAI Conference on Artificial Intelligence*, 2019.
- [10] Alistair E. W. Johnson, Tom J. Pollard, Nathaniel R. Greenbaum, Matthew P. Lungren, Chih-ying Deng, Yifan Peng, Zhiyong Lu, Roger G. Mark, Seth J. Berkowitz, and Steven Horng. MIMIC-CXR: A large publicly available database of labeled chest radiographs. *arXiv preprint arXiv:1901.07042*, 2019.
- [11] Pang Wei Koh, Shiori Sagawa, Henrik Marklund, Sang Michael Xie, Marvin Zhang, Akshay Balsubramani, Weihua Hu, Michihiro Yasunaga, Richard Lanus Phillips, Irena Gao, et al. WILDS: A benchmark of in-the-wild distribution shifts. In *Proceedings of the International Conference on Machine Learning (ICML)*, 2021.
- [12] Ananya Kumar, Percy Liang, and Tengyu Ma. Verified uncertainty calibration. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [13] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2017.

- [14] Yuan Liu, Ayush Jain, Clara Eng, David H. Way, Kang Lee, Peggy Bui, Kimberly Kanada, Guilherme de Oliveira Marinho, Jessica Gallegos, Sara Gabriele, et al. A deep learning system for differential diagnosis of skin diseases. *Nature Medicine*, 26:900–908, 2020.
- [15] Rachel Luo, Aadyot Bhatnagar, Yu Wang, Yu Bai, Liangliang Zhou, Mengyao Wang, Krzysztof Choromanski, Stefano Ermon Yu, and Marco Pavone. Local calibration: Metrics and recalibration. In *Proceedings of the Conference on Uncertainty in Artificial Intelligence (UAI)*, 2023.
- [16] Andrey Malinin and Mark Gales. Predictive uncertainty estimation via prior networks. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [17] David A. McAllester. PAC-Bayesian model averaging. In *Proceedings of the Twelfth Annual Conference on Computational Learning Theory (COLT)*, pages 164–170, 1999.
- [18] Bjoern H. Menze, Andras Jakab, Stefan Bauer, Jayashree Kalpathy-Cramer, Keyvan Farahani, Justin Kirby, Yuliya Burren, Nicole Porz, Johannes Slotboom, Roland Wiest, et al. The multimodal brain tumor image segmentation benchmark (BRATS). *IEEE Transactions on Medical Imaging*, 34(10):1993–2024, 2015.
- [19] Yaniv Ovadia, Emily Fertig, Jie Ren, Zachary Nado, D. Sculley, Sebastian Nowozin, Joshua V. Dillon, Balaji Lakshminarayanan, and Jasper Snoek. Can you trust your model’s uncertainty? Evaluating predictive uncertainty under dataset shift. *Advances in Neural Information Processing Systems (NeurIPS)*, 2019.
- [20] Andre G. C. Pacheco, Gustavo R. Lima, Amanda S. Salomão, Breno Krohling, Igor P. Biral, Gabriel G. de Angelo, Fábio C. R. Alves Jr, José G. M. Esgario, Alana C. Simora, Pedro B. C. Castro, et al. Pad-ufes-20: A skin lesion dataset composed of patient data and clinical images collected from smartphones. *Data in Brief*, 32:106221, 2020.
- [21] Seong Jin Park, Hyunwoo Kim, and Sangho Lee. Deep learning-based image quality assessment for dermoscopic images. In *IEEE International Symposium on Biomedical Imaging (ISBI)*, 2023.
- [22] Murat Sensoy, Lance Kaplan, and Melih Kandemir. Evidential deep learning to quantify classification uncertainty. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2018.
- [23] Laleh Seyyed-Kalantari, Haoran Zhang, Matthew B. A. McDermott, Irene Y. Chen, and Marzyeh Ghassemi. Underdiagnosis bias of artificial intelligence algorithms applied to chest radiographs in under-served patient populations. *Nature Medicine*, 27:2176–2182, 2021.
- [24] Adarsh Subbaswamy and Suchi Saria. Evaluating robustness of clinical AI models to distribution shift. In *Proceedings of the Conference on Health, Inference, and Learning (CHIL)*, 2021.
- [25] Philipp Tschandl, Cliff Rosendahl, and Harald Kittler. The ham10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions. *Scientific Data*, 5: 180161, 2018.
- [26] Philipp Tschandl, Christoph Rinner, Zoe Apalla, Giuseppe Argenziano, Noel Codella, Allan Halpern, Monika Janda, Aimilios Lallas, Caterina Longo, Josep Malvehy, et al. Human–computer collaboration for skin cancer recognition. *Nature Medicine*, 26:1229–1234, 2020.
- [27] Julia K. Winkler, Katharina Sies, Christine Fink, Ferdinand Toberer, Alexander Enk, Teresa Deindelein, Rainer Hofmann-Wellenhof, and Holger A. Haenssle. Non-invasive diagnosis of melanoma with artificial intelligence: A prospective study comparing smartphone-based and standard dermoscopy. *European Journal of Cancer*, 163:67–75, 2022.
- [28] Yaodong Yu, Stephen Bates, Yi Ma, and Michael I. Jordan. Robust calibration with multi-domain temperature scaling. In *Advances in Neural Information Processing Systems (NeurIPS)*, 2022.

-
- [29] Bianca Zadrozny and Charles Elkan. Transforming classifier scores into accurate multiclass probability estimates. In *Proceedings of the 8th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 694–699, 2002.